

# The Mechanism of Decay from the Bost–Connes System

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## Abstract

The structure, thermalization, and motion of matter have been established on the state space of the Bost–Connes system; the change of identity has so far had no mechanism. We define decay as the change of identity of a stable standing wave,  $\chi_j \rightarrow \chi_k$ , and it proceeds in only one way: the amplitude of the initial state merges into the background, the final state re-forms from the background, and an emitted pair carries the record away. Merging and formation are evaluated at the vertices against the background; the evaluation gives the conservation of electric charge, generation charge, and parity, the direction and rate of decay follow from it, and the Fermi constant and the muon lifetime are given by the numbers of the framework. The exponential decay law, the  $\beta$  spectrum, parity violation, the lifetime hierarchy, and nucleon stability are explained one by one from the same structure.

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# 1 The Definition of Decay

Matter is a stable standing wave maintained by cycles in a thermal bath[1]. Propagation repeats in units of cycles, and each cycle consists of three steps in sequence[1, 3]:

$$\dots \xrightarrow{\text{free evolution}} \cdot \xrightarrow{\text{formation exchange}} \cdot \xrightarrow{\text{pattern reconstruction}} \dots$$

Free evolution accumulates translation phase along the character direction; formation exchange trades one share of amplitude with the vacuum and is the only step of the cycle that communicates with the background; pattern reconstruction rebuilds the standing wave at a new center by stationary phase. The period is  $1/v$ , and the cycle rate  $v$  is shared by all excitations; mass is the phase per cycle multiplied by the cycle rate,  $m = v \times (\text{phase per cycle})$ , and exactly one share of phase is registered per cycle[1]. Propagation is carried by the convolution operator  $K$ , diagonal in the character basis, and changes of position are carried by the additive displacement  $T_a$ [1, 3].

Matter does not always propagate stably. A free neutron becomes a proton with a lifetime measured in minutes, and a muon becomes an electron after  $2.197 \times 10^{-6}$  seconds; no interaction can keep a neutron a neutron. The change nevertheless follows definite laws: the branch of the  $\tau$  decaying to a muon is always 0.1739, and the same particle decays at the same rate at any time and any place. What, then, is the cause of decay?

- **Why does decay exist?** Stability is maintained by detailed balance, the amplitude flowing out being taken back every cycle; within a self-consistently maintained structure, where does change come from?

- **What is the mechanism of decay?** Through what channel does the change of identity proceed, and by what rules is it constrained; which changes are allowed, which never occur, and what does the change carry away?
- **What determines the direction and rate of decay?** The spontaneous reverse process has never appeared; lifetimes extend from  $10^{-25}$  seconds to beyond  $10^{34}$  years, across more than sixty orders of magnitude. The uniqueness of the direction and the disparity of the hierarchy both require an origin.

## 1.1 Zero-Element Transit

Decay is a change of identity: a neutron becomes a proton, a muon becomes an electron. The species of a particle is carried by its character direction[1], and a change of identity is the transfer of a standing wave from one direction to another.

Of the three steps of the cycle, free evolution only accumulates phase and trades no amplitude with the background; pattern reconstruction rebuilds the standing wave within its direction by stationary phase; both steps take place inside the standing wave itself, and its form is unchanged in them. The only step that communicates with the background is formation exchange, in which the standing wave lets amplitude flow out and then takes it back[1]. A change of identity can occur only at formation exchange.

Identity is the form of the standing wave, and its change proceeds in two steps: the original species merges with its amplitude into the background, and the new species then forms from the background. Amplitude merged into the background belongs to no direction, so the position at which merging takes place must carry no species. Every character takes the value zero at the zero element, and the only position in the whole space that carries no species is the zero element, written 0. In the structure of positions the zero element is like any lattice site: the motion of a standing wave reaches it and leaves it; in the structure of species it is a vacancy: merged amplitude is not sent by propagation toward any direction, and merging therefore has no reverse. Motion through positions passes through the zero element as through any other site; species terminate at it. The zero element is the unique point at which the structure of positions and the structure of species disagree.

The operator form of the change follows. The derivation uses the section at the unit-group layer, the space  $L^2(\mathbb{Z}/p\mathbb{Z})$ , with positions labeled by the elements of  $\mathbb{Z}/p\mathbb{Z}$  and  $\delta_a$  the unit vector at position  $a$ . Changes of position are carried by addition and species by multiplication; the mathematical form of these two roles is the prescription of the two operations at the zero element: a displacement acts on the zero element as on any lattice site, addition treating 0 as an ordinary site;  $0 \cdot n = 0$  for all  $n$ , multiplication treating 0 as the absorbing element, with characters extended to the whole space by  $\chi(0) = 0$ . Addition and multiplication coexist compatibly on all nonzero sites; the zero element is the unique point at which the two operations disagree.

The section carries two families of basic operators: the convolution  $K$ , diagonal in the character basis, scales only along directions and preserves identity exactly; the displacement  $T_a$ ,  $(T_a f)(x) = f(x + a)$ , has the additive frequency modes  $e_\kappa(x) = p^{-1/2} e^{2\pi i \kappa x/p}$  as eigenvectors,

$$(T_a e_\kappa)(x) = e_\kappa(x + a) = e^{2\pi i \kappa a/p} e_\kappa(x).$$

The labels of identity are defined on the multiplicative character basis and the labels of motion on the additive frequency basis, and the two bases are Fourier duals of each other. The displacement is not diagonal in the character basis, and under it one direction develops nonzero components along the other directions; but a displacement is a unitary motion of the whole, emits no carrier,

and produces no record, and by the record condition[3] it does not constitute a process of identity change.

The operator form and the order of the change are fixed uniquely by the status of the zero element.

Step 1, compatibility with the multiplicative structure. The operator sought labels position; the zero element is a position, so the form is a diagonal operator on positions, a position function  $\theta(x)$ ; commuting with all multiplicative translations requires  $\theta$  to be constant on multiplicative orbits. The section has only two multiplicative orbits, the unit group and the zero element, so the compatible operators have the form

$$D = \theta_* \mathbb{K}_{(\mathbb{Z}/p\mathbb{Z})^*} + \theta_0 |\delta_0\rangle\langle\delta_0|.$$

Step 2, nontrivial commutation with displacements. The contribution of the identity component merges with the zero-element component;  $\theta_*$  is absorbed into the diagonal part of  $K$  and changes no direction; the nontrivial part is the zero-element component alone, normalized to  $\theta_0 = 1$ , with the overall strength absorbed into the cycle rate  $v$  (§4.3). Hence

$$D_F = |\delta_0\rangle\langle\delta_0|$$

Step 3, flat normalization. The additive Fourier expansion of  $\delta_0$  has equal weight on all frequencies,  $\langle e_\kappa | \delta_0 \rangle = p^{-1/2}$  for every  $\kappa$ , so the coupling of the transit to frequencies is flat; in §4 this guarantees that the forward rate is nonzero factor by factor.

Step 4, vanishing at first order and nonvanishing at second order. The commutator of the displacement with the projection is

$$[D_F, T_a] = |\delta_0\rangle\langle\delta_a| - |\delta_{-a}\rangle\langle\delta_0|,$$

whose two terms are absorption into the zero element and emission from it, recording exactly the two steps of merging and re-forming; if the two operations commuted, the commutator would vanish identically and no change would exist. A change of identity connects two directions, the initial standing wave along  $\chi_j$  and the final one along  $\chi_k$ , and the amplitude of the change is the matrix element of the change operator between them. The first-order matrix element

$$\langle \chi_k | [D_F, T_a] | \chi_j \rangle = \overline{\chi_k(0)} \chi_j(a) - \overline{\chi_k(-a)} \chi_j(0) = 0,$$

each term containing one value at the zero element, vanishes term by term since  $\chi(0) = 0$ : whatever the initial and final directions, a change completed in one step is identically zero. The composition of two steps is not zero. The first action goes through the absorption term, and the amplitude of the initial direction at position  $a$  enters the zero element; the second goes through the emission term, and the amplitude of the zero element is sent to position  $-b$ ; the composite matrix element is  $-\overline{\chi_k(-b)} \chi_j(a)$ . A character has modulus one at every site of the unit group, so this matrix element has modulus one identically: for any pair of directions  $\chi_j, \chi_k$ , the two-step transfer amplitude through the zero element is nonzero.

Zero at first order and nonzero at second order: this is the reason decay exists. Identity cannot be rewritten directly, yet between any two directions there is a two-step transfer amplitude through the zero element. Stability is the norm because of the first-order zero, and decay can occur because of the second-order nonzero; the two coexist in one structure.

A change of identity therefore passes through the zero element at the formation-exchange step and is completed by two vertices, absorption and emission, with one stay at the zero element: first merging into the background, then re-forming. This second-order process is called zero-element

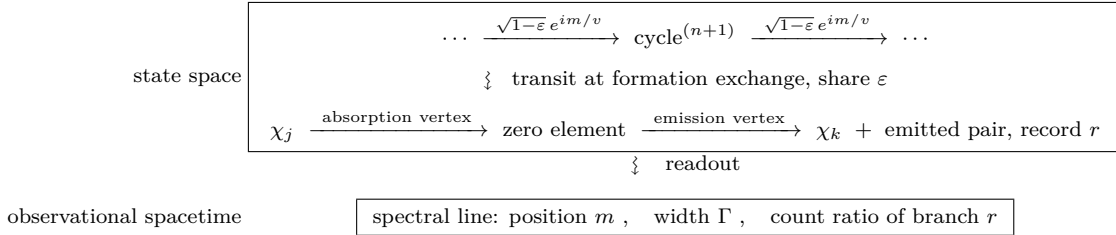
transit. Every decay passes through one zero-element transit; the exchange topology of two vertices and one transit is exactly the structure by which the weak interaction proceeds through  $W$  exchange[40]. The transit provides the existence of change without selection, the modulus of the composite matrix element being the same for every pair of directions; selection takes place at the two vertices at the ends of the transit.

## 1.2 The Elements of Decay

A process in the state space is read out through records; the structure of the readout is observational spacetime, in which the process appears as events[3]; the record criterion is applied to transit in §3.1. The picture of the cycle (§1) and the structure of transit (§1.1) together give the definition of decay in four elements.

- **Arena**—the arena of decay is the state space, the same as for formation, thermalization, and motion[1, 2, 3].
- **Object**—the object of decay is a stable standing wave maintained by cycles in the thermal bath, that is, matter itself.
- **Process**—the process of decay is transit through the zero element at the formation-exchange step, acting at most once in each cycle.
- **Record**—decay is completed when the lost amplitude turns into an escaping excitation carrying a record, whose readout is an event in observational spacetime (§3.1).

The complete structure of decay is



The upper box is the whole process in the state space: the cycles advance one by one, each cycle loses the share  $\varepsilon$ , the loss is distributed through the zero-element transit among the branches  $r$ , and the emitted pair escapes with the record; the lower box is the readout in observational spacetime, the position and width of the spectral line and the branch counts. Standard usage divides decays by interaction into strong, electromagnetic, and weak; the decay defined in this paper is the change of identity and corresponds to weak decay, while strong decay, electromagnetic decay, and  $\alpha$  decay do not change identity in this framework, and the classification of the three is given together in §6. Which losses are allowed and which forbidden is decided by the evaluation at the transit vertices, the subject of the next section.

## 2 The Mechanism of Decay

Zero-element transit establishes the existence of change, and the course of a transit is decided by the evaluation at its two vertices; the evaluation gives the rules of transit, and the rules determine the kinds of transit.

## 2.1 The Rules of Transit

Decay begins with a transit at the formation-exchange step. Of the three steps of the cycle only formation exchange trades amplitude with the background (§1.1); the transit is a second-order process on this step, acting at most once per cycle. One transit replaces one standing wave by a set of standing waves: the amplitude of the initial state merges into the zero element through the absorption vertex, the amplitude during the stay belongs to no direction, and each standing wave of the final state re-forms from the background through the emission vertex (§1.1). The completion of the change is judged by the record: the difference between the initial and final states must be carried away by a really emitted excitation (§3.1), so the final state contains at least two standing waves. Angular momentum is conserved event by event[3]; the standing waves of matter change sign under a  $2\pi$  rotation and have half-integer spin, one standing wave cannot be replaced by two, and the number of standing waves in the final state is odd; the standing waves emitted in addition to the line therefore come in pairs.

The standing waves taking part in a transit are identified as particle species by direction; the identification is fixed uniquely in reference [1], and the notation used in this paper is

| Direction        | Species                                      | Charge | Parity $\chi(-1)$ | Remarks                                                          |
|------------------|----------------------------------------------|--------|-------------------|------------------------------------------------------------------|
| $\chi_0$         | background; emergent states condense from it | 0      | even              | condensation direction of baryons                                |
| $\chi_1$         | up quark                                     | $+2/3$ | odd               | conjugate pair with $\chi_5$                                     |
| $\chi_2, \chi_4$ | charged lepton                               | $-1$   | even              | conjugate pair, the two winding directions of one standing wave  |
| $\chi_3$         | neutrino                                     | 0      | odd               | real direction; breaking direction, from which $W$ acquires mass |
| $\chi_5$         | down quark                                   | $-1/3$ | odd               | conjugate pair with $\chi_1$                                     |

In the CRT coordinates ( $k \bmod 2$ ,  $k \bmod 3$ ) of a direction, the  $\mathbb{Z}/2$  component is weak, the  $\mathbb{Z}/3$  component is color, and the cross-factor direction is electromagnetic; the lepton conjugate pair  $\{\chi_2, \chi_4\}$  carries the two winding directions of one species, while the quark conjugate pair  $\{\chi_1, \chi_5\}$  carries two species[1]. Electric charge is carried by direction and is conserved across interactions and transits[1].

The transit itself is equivalent for all directions (§1.1); the final state is decided by the vertices. Merging and formation are both evaluated at the vertices against the KMS background[1]; the final states retained by the evaluation constitute the allowed transits, and final states evaluating to zero do not occur. The rules of transit are given by the symmetry of the background. The symmetry compatible with both propagation and transit is the multiplicative action of the unit group on the lattice sites: the multiplication map  $\mu_c$  acts as a scalar in the character basis,

$$(\mu_c \chi_k)(x) = \chi_k(cx) = \chi_k(c) \chi_k(x),$$

fixes the zero element,  $\mu_c \delta_0 = \delta_0$ , and commutes with both the convolution operator  $K$  and the zero-element projection  $D_F$ . The unit group splits by CRT into two factors,  $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ , with  $\mathbb{Z}/2$  generated by  $\mu_{-1}$  and  $\mathbb{Z}/3$  by  $\mu_2$ ; each factor gives one conservation law through the vertex evaluation.

The first conservation law comes from  $\mu_2$ . The involution  $\iota$  pairs the lattice sites into doublets  $\{a, p - a\}$ , the position of a doublet is a generation, and there are three[1]; the doubling map  $\mu_2: a \mapsto 2a$  links the doublets in turn,

$$\{1, 6\} \xrightarrow{\mu_2} \{2, 5\} \xrightarrow{\mu_2} \{4, 3\} \xrightarrow{\mu_2} \{1, 6\},$$

the three generations form a three-cycle,  $\mu_2^3 = 1$  ( $2^3 = 8 \equiv 1 \pmod{7}$ ), the spectrum is  $\{1, \omega, \omega^2\}$  with  $\omega = e^{2\pi i/3}$ , and the three eigensectors give a  $\mathbb{Z}/3$  grading of the state space, whose label is called the **generation charge**. Characters are eigenfunctions of  $\mu_2$ , directions are not moved by  $\mu_2$ , and the generation charge is carried on the lattice side, established at the lattice readout and written into the record (§3.1).

Generation charge is conserved under propagation and transit; one transit changes it by at most one unit.

$K$  and  $\mu_2$  are simultaneously diagonal in the character basis,  $[K, \mu_2] = 0$ , and  $K$  preserves each eigensector; the vertices of  $K$  are scalar actions on directions, carry no lattice operator, and cannot rewrite an established generation record,  $\Delta g = 0$ .

$\mu_2$  fixes the zero element,  $\mu_2 D_F \mu_2^{-1} = |\mu_2 \delta_0\rangle\langle\mu_2 \delta_0| = D_F$ ,  $[D_F, \mu_2] = 0$ : a move of generation does not pass through the zero element and is not accompanied by emission, so a change of generation charge can occur only at the lattice projection[1] of the transit; the transit acts at most once per cycle, and a single transit moves at most one unit.

The two sides carry the change differently[1]. The inter-generation structure of the quarks contains a locked link, and the lattice projection reads out its share, the Cabibbo parameter  $\lambda$ [1]; between the generations of the leptons lie free lattice sites with no link to read, and a lepton keeps its generation in a transit. The generation charge flows from the initial state to one standing wave of the final state; that standing wave together with the initial state is called the **line**, and the remaining final standing waves are called the **emitted pair**, which is generation-neutral.

The second conservation law comes from  $\mu_{-1}$ . The involution  $\iota(a) = p - a$  is multiplication by  $-1$  and maps every direction to itself,

$$(\chi_k \circ \iota)(a) = \chi_k(-a) = \chi_k(-1) \chi_k(a),$$

with eigenvalue  $\chi_k(-1) = \pm 1$  the parity sign of the direction.  $-1$  is the unique element of order two in the unit group; with a primitive root  $g$ ,  $-1 = g^3$ ,  $\chi_k(g) = e^{2\pi i k/6}$ ,

$$\chi_k(-1) = e^{2\pi i \cdot 3k/6} = (-1)^k :$$

the charged leptons  $\chi_2, \chi_4$  are even, while the neutrino  $\chi_3$  and the two quarks  $\chi_1, \chi_5$  are odd; conjugate directions have the same sign,  $\chi_{-k}(-1) = (-1)^{-k} = (-1)^k$ ; composition corresponds to the product,  $(\chi\chi')(-1) = \chi(-1)\chi'(-1)$ , and the parity of a set of directions is the product of their signs. The total parity of a transit is the product of the signs of all initial and final directions times the product of the signs of the record components used.

The total parity of a transit is even.

Step 1, the record components. Even directions are invariant under  $\iota$ ; the lepton conjugate pair carries one species, and its identity record contains only even components; the quark conjugate pair carries two species, and the record component  $\eta$  distinguishing the two species satisfies  $\iota\eta = -\eta$ , an  $\iota$ -odd component: quarks are the unique source of odd components in records.

Step 2, the reflection of the transit. The transit matrix element

$$[D_F, T_a] = |\delta_0\rangle\langle\delta_a| - |\delta_{-a}\rangle\langle\delta_0|$$

absorbs at position  $a$  and emits at position  $-a$ ; passage through the zero element reflects the position  $a$  to  $-a$ , and the  $\iota$ -component of the record enters the evaluation with it. A quark line must change

species at the transit: the amplitude for keeping the species is proportional to the reflected overlap  $\langle ih|f\rangle$ , the two species are orthogonal, and this overlap vanishes identically.

Step 3, the evaluation by the background. Write the parity reflection as  $P$ ,  $(Pf)(x) = f(-x)$ ; then  $P\chi_k = \chi_k(-1)\chi_k$ ,  $P\delta_0 = \delta_0$ , and  $P[D_F, T_a]P = [D_F, T_{-a}]$ . The background is invariant under  $\iota$ , so the evaluation satisfies  $\varphi(A) = \varphi(PAP)$ ; for an operator that changes sign,  $PAP = -A$ ,  $\varphi(A) = \varphi(-A) = -\varphi(A)$ , hence  $\varphi(A) = 0$ . Only  $P$ -even combinations have nonzero evaluation, the total parity must be even, and this is the vertex-level form of the generation-projection identity in the appendix of reference [1]. Together: a transit whose direction parity is odd must involve at least one quark, whose odd record component restores evenness; a purely leptonic transit must have even direction parity by itself. The operator verification and the process-by-process check are in Appendix A.

The two conservation laws come from the two factors of the CRT decomposition, and there is no third symmetry compatible with both propagation and transit. The final state of one transit is fixed item by item by the three conservation laws of electric charge, generation charge, and parity; the kinds of transit sorted out by the conservation laws are the subject of the next section.

## 2.2 The Kinds of Transit

Once the three conservation laws are fixed, the final state of one transit and the organization of multiple transits are their corollaries.

Generation charge measures the change. Propagation does not change the generation charge (§2.1), so the change of generation charge along a path is contributed entirely by transits; the lattice projection of each transit moves at most one unit, the  $i$ th shift being  $\delta_i \in \{0, \pm 1\}$ , and the total change over  $n$  transits is

$$|\Delta g| = \left| \sum_i \delta_i \right| \leq \sum_i |\delta_i| \leq n.$$

A change spanning  $|\Delta g|$  generations needs at least  $|\Delta g|$  transits: a change of at most one generation is completed in a single transit, and one of more than one generation necessarily takes several.

The organization of multiple transits is decided by whether each one is completed. A completed transit is accompanied by emission, and the carrier takes the difference of that transit away and escapes without return (§2.1); once the difference has been carried away, successive transits become distinguishable events, distinguishable degrees of freedom are counted at the probability level[1], and the shares of the transits multiply. An uncompleted transit has no emission and is a virtual process; between virtual processes no carrier tells which came first, they are indistinguishable, indistinguishable degrees of freedom are normalized at the amplitude level[1], and the matrix elements compose in amplitude. These are the only two ways of joining transits, so decay proceeds in exactly three modes: a single transit, a cascade of completed transits, and an amplitude-level composition containing virtual processes.

### 2.2.1 Single Transit

The final state of one transit is fixed item by item by electric charge, generation charge, and parity.

Charge restricts the emitted pair. The emitted pair forms from the background, and its charge is carried by the channel excitation through which the transit passes; in the boson spectrum of reference [1], channel excitations carry only the charges  $\pm 1$  ( $W$ , the breaking direction  $\chi_3$ ) and 0 ( $Z$ , photon, gluon), there is no direction carrying fractional charge, and the charge of an emitted pair is therefore 0 or  $\pm 1$ . The admissible emitted pairs are thus of two kinds only:



- lepton pairs:  $(\ell^-, \bar{\nu}_\ell)$  and  $(\ell^+, \nu_\ell)$  carrying  $\mp 1$ ,  $(\ell, \bar{\ell})$  and  $(\nu, \bar{\nu})$  carrying 0;
- quark pairs:  $(u, \bar{d})$  and  $(d, \bar{u})$  carrying  $\pm(2/3 + 1/3) = \pm 1$ ,  $(q, \bar{q})$  carrying 0.

A mixed pair of one quark and one lepton carries fractional charge and does not occur.

Charge restricts the line. The charge difference between the two ends of the line equals the charge of the emitted pair and must be an integer; the charge differences between a quark and a charged lepton or a neutrino are  $\pm 1/3$ ,  $\pm 2/3$ , or  $\pm 5/3$ , all fractional, so there is no transit between quarks and leptons. A change within one species does not change identity. Changes of identity are therefore only of two kinds,

$$\boxed{d \leftrightarrow u, \quad \ell \leftrightarrow \nu_\ell}$$

each with a charge difference of one unit, carried by the emitted pair.

Generation charge fixes the generation. A lepton keeps its generation in a transit (§2.1); in  $\ell \rightarrow \nu_\ell$  the neutrino is of the same generation as the initial state, the generation charge leaves with the neutrino, and the emitted pair is generation-neutral: the final state of a  $\mu$  must contain  $\nu_\mu$ , and that of a  $\tau$  must contain  $\nu_\tau$ . The change  $\mu \rightarrow e$  between two charged leptons would have to move one unit of generation charge, the lepton side has no lattice projection to carry it, and the amplitude is zero. A quark transit may stay in its generation or shift by one:  $d \rightarrow u$  stays,  $s \rightarrow u$  shifts by one generation, the lattice projection carries the share  $\lambda$ , and the probability-level suppression is  $\lambda^2 = 0.0505 \approx 1/20$ .

Parity fixes the pairing of line and emitted pair. The direction parity of the line  $d \rightarrow u$  is  $(-)(-) = +$ . Paired with the lepton pair  $(\ell^-, \bar{\nu}_\ell)$ , whose parity is  $(+)(-) = -$ , the total parity is odd, restored to even by the odd record component of the quark on the line: allowed. Paired with the quark pair  $(d, \bar{u})$ , whose parity is  $(-)(-) = +$ , the total parity is even: allowed.

The direction parity of the line  $\ell \rightarrow \nu_\ell$  is  $(+)(-) = -$ . Paired with the lepton pair  $(\ell'^-, \bar{\nu}_{\ell'})$ , of odd parity, the total parity is even: allowed. Paired with the quark pair  $(d, \bar{u})$ , of even parity, the total parity is odd, restored to even by the odd record component of the quark in the emitted pair: allowed. Paired with  $(\ell, \bar{\ell})$  or  $(\nu, \bar{\nu})$ , of even parity, the total parity is odd with no quark involved: zero.

The kinds of single transit are thereby exhausted, exactly four:

- quark line with lepton pair,  $d \rightarrow u + (\ell^-, \bar{\nu}_\ell)$ :  $\beta$  decay;
- quark line with quark pair,  $d \rightarrow u + (d, \bar{u})$ : nonleptonic decay;
- lepton line with lepton pair,  $\ell \rightarrow \nu_\ell + (\ell'^-, \bar{\nu}_{\ell'})$ : the decay of the  $\mu$  and the leptonic branches of the  $\tau$ ;
- lepton line with quark pair,  $\ell \rightarrow \nu_\ell + (d, \bar{u})$ : the hadronic branches of the  $\tau$ .

The reverse of each ( $u \rightarrow d$ ,  $\nu_\ell \rightarrow \ell$ ) is equally allowed at the vertex level; whether it occurs is decided by the completion condition of the final state (§2.3). The emitted pairs of the  $\tau$  are  $(e, \bar{\nu}_e)$ ,  $(\mu, \bar{\nu}_\mu)$ , and  $(d, \bar{u})$  carrying three colors, with branch counts 1 : 1 : 3. The parities of the four pairings are checked process by process in Appendix A.

$W$  and  $Z$  occupy different positions in this structure.  $W$  belongs to the breaking direction  $\chi_3$  and is odd; each appearance of it is the two-vertex structure of a transit: the decay of  $W$  is the emission of a transit, obeys parity conservation, and its final state is necessarily one of the two kinds of emitted pair.  $Z$  belongs to an even direction and is a channel excitation; the excitation returns its weight to the channel without changing the identity of any standing wave, which is called **release**, and passes through no transit; both the visible and the invisible final states are open, and the structure of the width is given in reference [1].

### 2.2.2 Multiple Transits

Multiple transits divide into two kinds by the level at which they are joined.

The joining of completed transits is a cascade. Each transit is accompanied by its own emission (§2.1), the carrier takes the difference of that transit away, successive transits are mutually distinguishable events, and the probability of reaching a specified final state along the chain is the product of the probabilities,

$$\mathcal{P}(\text{chain}) = \prod_i \mathcal{P}_i.$$

A cascade introduces no new kind; each transit in the chain is one of the four. A change of more than one generation proceeds by cascade: a chain with  $|\Delta g| = n$  takes exactly  $n$  transits, each shifting one generation and each carrying  $\lambda$ , with probability-level suppression  $\lambda^{2n}$ ; leptons cannot move generation (§2.1), so cross-generation cascades are carried entirely by quarks. The final state of a heavier state after one decay may decay again, stepping down until stable matter is reached; cascades of this kind exist on both the quark and the lepton side. A recorded cascade cannot return; the chain only advances.

The joining that contains virtual processes is amplitude-level composition. Between the transits there is no emission and no record, the transits are indistinguishable, and the total amplitude is the product of the matrix elements,

$$A(\text{composition}) = \prod_i A_i.$$

The mixing of  $K^0$  and  $\bar{K}^0$  is this process[36]: the two lines  $d \rightarrow s$  and  $\bar{s} \rightarrow \bar{d}$  each span one generation and each pass through one transit, both transits are virtual, the four vertices compose into one amplitude-level process, the mixing amplitude is of the order of the square of a single amplitude, and the smallness of the mixing follows. An unrecorded composition is reversible, so  $K^0$  and  $\bar{K}^0$  oscillate back and forth, in contrast to the irreversibility of a cascade.

The kinds of transit are thereby exhausted. Changes of identity are only  $d \leftrightarrow u$  and  $\ell \leftrightarrow \nu_\ell$ , emitted pairs are only lepton pairs and quark pairs, cross-generation changes proceed by cascade, and unrecorded compositions appear only at the amplitude level; the decay of emergent states is the subject of the next section.

## 2.3 Decay Branches

The kinds of transit give the changes allowed at the vertex level; for a change to become a decay, the final state must also be completable: every standing wave of the final state has to complete its own formation cycle, so the sum of the final masses is smaller than the initial mass. The reverses of the four transits ( $u \rightarrow d$ ,  $\nu_\ell \rightarrow \ell$ ) have heavier final states and are closed by this. The allowed transits together with the completable final states give all the decay branches and the stable matter.

The condensate body of an emergent state is maintained by the pole, does not cycle, and the reading of complex energy (§3.3) does not apply to it directly[1]. A baryon is not a bound state of three quarks but a single collective condensation in the  $\chi_0$  direction; the readings of its constituents are internal projections along the nontrivial character directions, and the shares of  $\chi_1$  and  $\chi_5$  are the up- and down-quark constituents[1]. The decay of an emergent state is therefore a transit of an internal projection, and the conservation laws and the completion condition of the preceding two sections apply item by item. The internal projection of the neutron has exactly one open branch,  $d \rightarrow u$  (§2.2); the neutron-proton mass difference  $\Delta m = 1.30 \text{ MeV}$  exceeds  $m_e$ [1], and phase space remains after the electron is completed; for the proton the same branch runs in reverse, the final state cannot be completed, and the branches in which a quark becomes a lepton or a neutrino are

closed by charge (§2.2). Emergent and elementary states are unified in the mechanism of decay; the magnitude of the neutron decay is given in §5.5.

The three conservation laws, the three modes, and the completion condition give the complete table of branches at  $p = 7$ :

| Initial state                   | Final state                                                           | Identity change                                     | Mode                                                   |
|---------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| $n$                             | $p e^- \bar{\nu}_e$                                                   | $d \rightarrow u$ , same generation                 | single; unique branch                                  |
| $\mu^-$                         | $e^- \bar{\nu}_e \nu_\mu$                                             | $\mu \rightarrow \nu_\mu$                           | single; unique branch                                  |
| $\tau^-$                        | $\nu_\tau + \{e \bar{\nu}_e, \mu \bar{\nu}_\mu, d \bar{u} \times 3\}$ | $\tau \rightarrow \nu_\tau$                         | single; three kinds, counts 1 : 1 : 3                  |
| $s \rightarrow u$ type          | strange decays                                                        | $s \rightarrow u$ , shift of one generation         | single; share $\lambda^2 \approx 1/20$                 |
| heavy-state chains              | stepwise downward                                                     | multiple                                            | cascade; probabilities multiply                        |
| $K^0 \leftrightarrow \bar{K}^0$ | mixing oscillation                                                    | $d \rightarrow s$ and $\bar{s} \rightarrow \bar{d}$ | amplitude-level composition; four vertices, reversible |
| $\pi^\pm$                       | $\ell \bar{\nu}_\ell$                                                 | $u \rightarrow d$ , annihilating with $\bar{d}$     | single; parity seed, $e$ channel suppressed (§4.7)     |
| $W$                             | $\ell \bar{\nu}_\ell \times 3, q \bar{q}' \times 6$                   | emission of a transit                               | single; nine branches, counts 3 : 6                    |
| $Z$                             | visible and invisible                                                 | no identity change                                  | resonance release; bypasses the zero element           |

Every row of the table is sorted out uniquely by the conservation laws (§2.1), the kinds (§2.2), and the completion condition.

States with all branches closed constitute stable matter.

- The proton. Its condensate body is the lowest color singlet, the condensate layer has no lower final state, and the constituent directions have no  $\chi_0$  matrix element[1]; among the branches of the internal projections, those in which a quark becomes a quark have heavier final states, and those in which a quark becomes a lepton or a neutrino have fractional charge differences; the two kinds are closed separately.
- The electron. The lightest charged stable state; under charge conservation there is no charged final state of lower mass.
- The neutrino. The lightest fermion, electrically neutral, with no insertion flow[1].
- The photon. Massless, carrying no cycle, with no share to lose.

Each forbidden process has a definite source of closure:  $\mu \rightarrow e \gamma$  is closed by the conservation of generation charge,  $\mu \rightarrow 3e$  by the absence of links between lepton generations, emitted pairs of type  $(q, \bar{\ell})$  by charge conservation, all branches of the proton by the completion condition and charge, and  $n-\bar{n}$  oscillation by the fact that all colored directions are complex characters[1]. The allowed changes and the closed states come from the same set of vertex identities. By what laws the branches proceed is the subject of the next section.

### 3 The Records of Decay

A transit replaces one standing wave by a set of standing waves (§2), and the process takes place in the state space; for it to become an event in observational spacetime it must be recorded (§1.2). The record criterion applied to transit gives the condition for a transit to become an event, the counting of events, and the readings of an event.

### 3.1 The Record Criterion

A record is given by the conjunction of three conditions[3]: the recorded difference is carried by classical quantum numbers commuting with propagation; the difference is carried away by a really emitted excitation; the carrier escapes without return. Each of the three takes a definite form on the transit.

The first condition, the carrying quantities. Propagation is carried by the convolution operator  $K$ , diagonal in the character basis,  $K\chi_k = L_k\chi_k$  with  $L_k = L(\beta, \chi_k)[1]$ ; write  $\Pi_k = |\chi_k\rangle\langle\chi_k|$  for the projection onto the direction  $\chi_k$ . The difference between initial and final states is carried by three quantities. Electric charge takes its value by direction (the table of §2.1),

$$Q = \sum_k q_k \Pi_k, \quad QK = \sum_k q_k L_k \Pi_k = KQ,$$

$[Q, K] = 0$ . Generation charge is the label of the eigensectors of the doubling map  $\mu_2$ , and the sector projections are polynomials in  $\mu_2$ ,

$$\Pi_g = \frac{1}{3} \sum_{j=0}^2 \omega^{-gj} \mu_2^j, \quad g \in \mathbb{Z}/3, \quad G = \sum_g g \Pi_g,$$

and  $[K, \mu_2] = 0$  (§2.1) gives  $[K, \Pi_g] = 0$ , hence  $[G, K] = 0$ . The parity reflection  $P$  is diagonal in the character basis,  $P\chi_k = (-1)^k \chi_k$  (§2.1),  $P = \sum_k (-1)^k \Pi_k$ ,  $[P, K] = 0$ .

The three are unchanged in propagation and take definite values at the lattice readout, charge in  $\{0, \pm 1/3, \pm 2/3, \pm 1\}$ , generation charge in  $\mathbb{Z}/3$ , parity in  $\pm 1$ ; the effective overlap of a record takes only the values 0 and 1[1, 3], and all three are classical quantum numbers. The winding direction is not among them: the readout takes Galois-conjugation-invariant combinations[1], the two winding directions of a conjugate pair are indistinguishable in the readout, the difference of winding direction does not enter the record, and the identity record of a lepton therefore contains only even components (§2.1). The differences of the three quantum numbers across the transit are taken up by the final state: the charge difference is carried by the emitted pair, the generation charge continues along the line or moves by one unit, and the product of parities is even (§2.2).

The second condition, emission. The difference must be carried away by a really emitted excitation. Each standing wave of the final state forms from the background at the emission vertex; formation is the repeated application of propagation and is completed when the standing wave has built its own cycle[1]; the mass of each final standing wave is read from its cycle, and the condition for formation to be completed is that the energy of the initial state suffices to complete all final states,

$$\sum_{\text{out}} m < m_{\text{in}},$$

the completion condition of §2.3. A completed standing wave propagates by its own cycle[3] and is a real emission; the record of an elastic process is carried by a photon[2], and the record of a decay is carried by the emitted standing waves themselves. When the final state cannot be completed, emission fails, the transit is not completed, and it enters only the amplitude (§2.2). The reality of the emission and the completion of the final state are one and the same condition.

The third condition, irreversibility. The carrier escapes without return. The completed final standing waves leave the emission vertex along their own cycles, and their overlap with the source decays to zero under propagation[3]; a return would require the emitted pair to become coherent again with the source at the same vertex, the overlap is zero, and the record cannot be undone. The structural origin of the absence of return is the zero element: amplitude merged into the zero element

is not sent by propagation toward any direction, and merging has no reverse (§1.1), developed in §4.

The conjunction of the three conditions is the completion of the transit:

The record of a decay consists of the differences of charge, generation charge, and parity carried by the final standing waves; with the final state completed and the carrier escaped the transit is complete, and lacking either it is a virtual process.

A completed transit is read out in observational spacetime as one event; the counting and the readings of events are the subject of the next two subsections.

### 3.2 Records and Events

A completed transit is read out in observational spacetime as one event; an uncompleted transit is a virtual process and has no event. One event carries three items:

- the recorded difference, that is, the branch  $r$ : the kind of the emitted pair and the shift of generation charge;
- the cycle number  $n$  at which it occurred;
- the energies and momenta of the final standing waves, carried by the phase space of the final state (§4).

Events are counted in units of cycles. The transit acts at most once per cycle (§2.1); the probability of completion is written  $\varepsilon$  and called the **loss share**;  $\varepsilon$  is fixed by the vertex evaluation, is constant from cycle to cycle, and its value is given in §4. Survival has no record and proceeds at the amplitude level: the survival amplitude of one cycle is the cycle eigenvalue

$$\Lambda = \sqrt{1 - \varepsilon} e^{im/v}$$

(the structure diagram of §1.2), the survival amplitude over  $n$  cycles is  $\Lambda^n$ , and the survival probability is  $|\Lambda|^{2n} = (1 - \varepsilon)^n$ . The probability that the event occurs in the  $n$ th cycle is

$$\mathcal{P}(n) = (1 - \varepsilon)^{n-1} \varepsilon, \quad \sum_{n \geq 1} \mathcal{P}(n) = 1, \quad \bar{n} = \frac{1}{\varepsilon},$$

a standing wave with  $\varepsilon > 0$  eventually decays, with mean lifetime  $1/\varepsilon$  cycles; the distribution depends only on  $\varepsilon$  and not on the number of cycles already survived, and the event has no memory (§5.1).

Events of different branches carry different records, are mutually distinguishable events, and add at the probability level,

$$\varepsilon = \sum_r \varepsilon_r,$$

with  $r$  running over the branches of the initial state in the table of branches (§2.3). Virtual processes do not count toward  $\varepsilon$ : an uncompleted transit has no record, its amplitude enters the surviving standing wave and belongs to the correction layer[1], accounted for in §4.4. The two levels of the readout are thereby fixed: events are counted at the probability level and survival proceeds at the amplitude level[1]; the three readings of an event are the subject of the next subsection.

### 3.3 The Readings of an Event

An event has three kinds of reading, the width of the spectral line, the position of the spectral line, and the count ratio of the branches (the structure diagram of §1.2), each derived from the cycle.

The width comes from the counting of events. Convert the survival probability  $(1 - \varepsilon)^n$  to time: one lattice step per cycle, of length  $1/v$ , the same for all standing waves[3], so  $n = vt$ . Survival probabilities multiply over successive intervals,  $S(t_1 + t_2) = S(t_1) S(t_2)$ , and the monotone solution is the exponential,

$$N(t) = N_0 (1 - \varepsilon)^{vt} = N_0 e^{-\Gamma t}, \quad \Gamma = -v \log(1 - \varepsilon) = v \varepsilon + O(\varepsilon^2),$$

so the exponential decay law[5] is a theorem. The shape of the line is given by the resolvent of the sum over cycles, the Fourier sum over the cycle count at energy  $E$ ,

$$A(E) = \sum_{n \geq 0} \Lambda^n e^{-iEn/v} = \frac{1}{1 - \Lambda e^{-iE/v}},$$

convergent since  $|\Lambda| < 1$ . Near  $E \approx m$ ,  $\Lambda e^{-iE/v} = \sqrt{1 - \varepsilon} e^{i(m-E)/v} \approx 1 - \varepsilon/2 + i(m - E)/v$ ,

$$A(E) \approx \frac{v}{\Gamma/2 + i(E - m)}, \quad |A(E)|^2 \propto \frac{1}{(E - m)^2 + \Gamma^2/4},$$

the Breit–Wigner distribution[9]; the line center reads the mass, and the line width reads the lifetime.

The mass comes from the surviving amplitude. Survival is at the amplitude level; the modulus of the eigenvalue  $\Lambda$  is given by  $\varepsilon$  and its phase by the phase per cycle  $m/v$  (§1, the frequency-readout lemma[1]); taking the logarithm and multiplying by the cycle rate,

$$v \log \Lambda = i m - \frac{\Gamma}{2},$$

the imaginary part is the mass and the real part is half the width; the one half in front of  $\Gamma$  comes from the difference between the two levels, loss being counted at the probability level while the surviving amplitude bears only half of the logarithm. The two readings come from one eigenvalue, the complex energy

$$\omega = -iv \log \Lambda = m + \frac{i\Gamma}{2}$$

being the pole of the resolvent[31]; with energies written as  $e^{-iEt}$  one takes the conjugate,  $E = m - i\Gamma/2$ , the usual form of the Breit–Wigner distribution. The domain condition of the mass readout,  $m < \pi v[1]$ , and the condition of the width readout,  $\varepsilon \ll 1$ , are independent of each other; the modulus has no lattice multivaluedness, and the width adds no new domain condition.

The count ratio of the branches comes from the distribution of the loss. Branches are mutually distinguishable events (§3.2),

$$\text{BR}_r = \frac{\varepsilon_r}{\varepsilon},$$

a branching ratio is a ratio of shares and a structural quantity. The share of each branch factorizes into three parts: the evaluation of the two vertices (strength absorbed into  $v$ ), the count of the branch, and the phase space of the final state; the phase space is a power of  $(m/v)$ , the exponent equal to the number of free final-state energies, and its continuous form is given by the inheritance of the spacetime structure (§4.2).

The three readings are now complete: the width from counting, the mass from survival, and the branching ratio from distribution, all given by the cycle eigenvalue and the distribution of the loss. Why decay runs in one direction only is the subject of the next section.

## 4 The Rate of Decay

The rules, kinds, and records of transit are established; what remains is the third question of the first section, what determines the direction and rate of decay. The loss runs in one direction only, and the loss share per cycle is converted into a continuous rate; the evaluation of rates proceeds in four layers, the formula, the calibration by the muon, the dwell correction for branches containing quarks, and the application case by case.

### 4.1 The Irreversibility of Decay

The direction of time has been established separately at the three levels of structure, ensemble, and event[1, 2, 3]; the one-way character of decay belongs to the rate of a single process. The dynamics itself is time-reversal symmetric:  $\sigma_t$  is unitary and invertible, and the transit matrix element satisfies under conjugation

$$[D_F, T_a]^\dagger = -[D_F, T_{-a}],$$

so the forward and reverse vertices have equal modulus, and the asymmetry does not come from the vertices.

The asymmetry appears in the rates. On the forward side, the loss share is nonzero factor by factor,

$$\varepsilon = 2\pi \sum_r |a_r|^2 \tilde{\rho}_r,$$

with  $a_r$  the vertex amplitude of branch  $r$  and  $\tilde{\rho}_r$  the density of final states (§4.2): flat normalization is a theorem, the transit coupling to every frequency with equal weight (§1.1); admissible branches exist (§2.2); the density of states is positive when the final state can be completed (§2.3). None of the three factors vanishes, and  $\Gamma > 0$ . On the reverse side, the rate of a spontaneous reverse process is proportional to the occupation on the incoming side; after the loss the local occupation is zero, and by three facts it does not rise again: thermal activation vanishes at zero temperature, vacuum freezing[2]; the carrier escapes without return, a controlled reversal would require gathering the escaped carriers, and this is excluded by the replacement of the environment[2] (§3.1); the dilation space is infinite-dimensional, and the occupation has no Poincaré recurrence[2]. The escape of the carrier cannot be undone, and the reverse rate is identically zero.

The one-way character is borne algebraically by the zero element. Addition is a group, every displacement has an inverse,  $T_a T_{-a} = \mathbb{K}$ , and a change of position can be retraced; multiplication is a semigroup, the zero element absorbs everything,  $0 \cdot n = 0$  for all  $n$ , and no multiplication sends the zero element back to a nonzero element. The zero element is the unique point at which the two operations disagree, and the existence of decay comes from it (§1.1); the one-way character of decay comes from the one-sidedness of the absorbing element: all other links of the process are reversible, propagation, displacement, and the two mutually conjugate vertices, and the only one-sided link is that the zero element absorbs and does not release. The zero element has no inverse under multiplication, and the loss in decay has no return flow: both come from one semigroup property.

Irreversibility does not belong to the dynamical equation but to the boundary between the process and its environment: the equation is reversible, the boundary one-way. The two facts involving the environment come from the replacement of the environment in reference [2], and the third is the dimension of the dilation space; the reverse vertices exist with equal modulus, and only the spontaneous reverse process is excluded. The relation to thermalization is fixed thereby: thermalization relaxes through the zero element at the ensemble level, and the change of identity is the slowest tier of the relaxation catalogue[2]; decay changes identity through the same zero element at the level of a single system, proceeds even at zero temperature, is a coherent process of

the dilation layer, and its rate is given by the modulus deficit of the eigenvalue with no temperature factor. The direction is established; the numerical values of the rates are the subject of the following subsections.

## 4.2 The Rate Formula

The cycle map with transit is

$$M = \mathcal{U}(\mathbb{K} - \mathcal{E}_{\text{form}} - \mathcal{E}_{\text{transit}}),$$

the first two terms maintaining the standing wave (§1) and the third being the loss; a decaying state is an eigenmode of  $M$ . The sum over cycles is the geometric series of the resolvent,

$$\sum_{n \geq 0} \Lambda^n = \frac{1}{1 - \Lambda},$$

the same mechanism as the sum over self-loops in the mass-evaluation rules of reference [1]; the sum of the series has its pole at the complex energy of §3.3 ( $E = m - i\Gamma/2$  in the  $e^{-iEt}$  convention), and energy conservation is an output of the summation. The modulus deficit per cycle at the pole is

$$\varepsilon = 2\pi \sum_r |a_r|^2 \tilde{\rho}_r,$$

the golden rule in discrete time:  $a_r$  is the vertex amplitude of branch  $r$ , dimensionless;  $\tilde{\rho}_r$  is the density of states per unit of the dimensionless energy  $E/v$ ,  $\tilde{\rho}_r = v\rho_r$ . The strength of the transit is absorbed into  $v$  (§4.3), the contact matrix element is  $M_r = v a_r$ , and  $\Gamma = v\varepsilon = 2\pi \sum_r |M_r|^2 \rho_r$  is the golden rule in continuous time. On returning to relativistic normalization, the normalization  $\sqrt{2E}$  of each final line, the flux factor  $1/(2m)$ , and the Lorentz-invariant phase space are all inherited from the spacetime structure[1]:

$$\Gamma = v\varepsilon = \frac{1}{2m} \int |M|^2 d\text{LIPS}$$

The time integral appears only once, in the sum over cycles, with no double counting. The duration of a single cycle,  $1/v$ , is far shorter than  $1/m$ , and the transition probability of a single cycle lies in the quadratic regime;  $\varepsilon$  is defined as the modulus deficit of the eigenvalue per cycle rather than as a short-time transition probability, and the Zeno difficulty[41] does not arise. The formulas below are the integrals, over the inherited phase space, of the contact matrix element of the framework (the  $V-A$  pairing of §2.1, with the spin structure from reference [3]); the results of the integration agree with the standard forms[12].

## 4.3 The Muon Lifetime

The strength of the transit is absorbed into the cycle rate  $v$  (the normalization of §1.1), which fixes the coupling of the contact form; the experimental definition of  $G_F$ ,  $v = (\sqrt{2} G_F)^{-1/2}$ , then gives

$$G_F = \frac{1}{\sqrt{2} v^2} = 1.16665 \times 10^{-5} \text{ GeV}^{-2},$$

deviating from the experimental value[42] by +0.023%. The unique branch of the muon gives

$$\Gamma_\mu = \frac{G_F^2 m_\mu^5}{192\pi^3} f\left(\frac{m_e^2}{m_\mu^2}\right), \quad f(x) = 1 - 8x + 8x^3 - x^4 - 12x^2 \log x,$$



with  $f$  the phase-space integral of the three-body final state and the masses taken as the spectral values computed in reference [1]. Numerically

$$\tau_\mu = 2.18709 \times 10^{-6} \text{ s},$$

a deviation of  $-0.45\%$ [42]. The residual is attributed to the electromagnetic correction of the charged lepton lines: the dwell of the two charged lines,  $\mu$  and  $e$ , through the electromagnetic channel is not derived in this paper; in the standard calculation this share is  $\frac{\alpha}{2\pi} \left( \frac{25}{4} - \pi^2 \right) = -0.42\%$ [15], comparable to the residual. The calibration is complete; branches containing quarks require in addition the dwell correction, the subject of the next subsection.

#### 4.4 The Dwell Correction

The residence of a lepton (its character direction[1]) forms in one step from a single generator, and the record is established as soon as it is emitted; the dwell of a charged lepton line with the electromagnetic channel passes through no broken segment, and its form is not derived in this paper (§4.3). The formation of a quark pair passes through the color channel and has a stretch of dwell, its endpoints not closed and its labels not fixed, and the correction lives on this broken segment (the stretch of the formation path on which the record is not yet established[1]).

Step 1, exactly one composite per segment. The correction of a broken segment is a two-vertex composite across the segment; the transition side of the segment is the passage through the zero element, instantaneous and without dwell, and a repeated composite has no place on the same segment: each broken segment carries exactly one composite[1].

Step 2, the catalogue of insertion channels. A channel taking part in the composite must have a massless carrier, that is, color or electromagnetism. The weak channel is broken, the carrier mass confines it within the subnuclear range [1, 4], and it provides no massless insertion; the change of identity itself is carried by the zero-element transit (§1.1). The catalogue has two entries only.

Step 3, configurations and shares. The configuration space of the composite consists of the combinations of the insertion-pair member and the line; the normalization is shared equally over the unitary displacement algebra, independent of the channel charge. The vertex coupling is proportional to the channel charge, and a color-neutral and electrically neutral intermediate residence (the direction  $\chi_3$ ) makes the contribution of half of the configurations zero, the generalization of the electric-neutrality exemption[1]: the share of nonzero configurations for each kind of composite is  $\frac{1}{2}$ . The configurations are counted one by one in Appendix B.

Step 4, the counting of labels. The generators of a color composite carry color labels, and the intermediate state sums over the  $d_2 = 3$  color labels released; the generators of an electromagnetic composite carry no color label, and the intermediate state is a single term. The total share of a single composite is

$$\xi = \frac{1}{2} d_2 \alpha_s^{(0)} + \frac{1}{2} \alpha_{\text{EM}}^{(0)} = 0.176343,$$

where  $\alpha_s^{(0)} = |R_1|^2/3 = 0.11492$  and  $\alpha_{\text{EM}}^{(0)} = |R_2|^2/6 = 0.0079270$  are the Born-level couplings: couplings of real processes take corrected-level observed values and inputs inside the correction layer take Born-level values (the evaluation rule of reference [1]); they are built from the resonance ratios  $R_k$  computed in reference [1].

Step 5, nesting. The intermediate state of the composite is itself a broken segment: the intermediate pair propagates in dwell through the color channel between the two vertices, its endpoints not closed and its labels not locked, agreeing item by item with the definition of a broken segment. By Step 1 it carries exactly one composite of its own; the recursion terminates only at a record,

the shares of the layers multiply, and the sum has the same form as the resolvent of the sum over cycles (§4.2):

$$1 + \xi + \xi^2 + \dots = \frac{1}{1 - \xi}, \quad \boxed{w_{\text{had}} = \frac{d_2}{1 - \xi} = 3.642}$$

The form of the correction is a nested resolvent, layer upon layer; the catalogue of insertion channels is the same at every layer, because the object of the recursion is the segment, and the properties of a segment do not depend on the kind of composite occupying it.

## 4.5 $\tau$ Decay

The weights of the three kinds of  $\tau$  branch are  $f_e : f_\mu : w_{\text{had}}$ , where  $f_\ell = f(m_\ell^2/m_\tau^2)$  is the phase-space factor,  $f_e = 0.9999993$ ,  $f_\mu = 0.972564$ ; the correction of the hadronic branch is given by the dwell (§4.4), and the total width is

$$\Gamma_\tau^{\text{tot}} = \frac{G_F^2 m_\tau^5}{192\pi^3} [f_e + f_\mu + w_{\text{had}}],$$

with  $m_\tau$  the spectral value computed in reference [1]. The numbers compared with experiment:

| Quantity                                                  | This paper                | Experiment                | Deviation |
|-----------------------------------------------------------|---------------------------|---------------------------|-----------|
| $\Gamma(\tau \rightarrow e\bar{\nu}_e\nu_\tau)$ [GeV]     | $4.050 \times 10^{-13}$   | $4.040 \times 10^{-13}$   | +0.25%    |
| $\Gamma(\tau \rightarrow \mu\bar{\nu}_\mu\nu_\tau)$ [GeV] | $3.939 \times 10^{-13}$   | $3.943 \times 10^{-13}$   | −0.10%    |
| $\text{BR}(\tau \rightarrow \mu\bar{\nu}_\mu\nu_\tau)$    | 0.1732                    | 0.1739                    | −0.40%    |
| $\tau_\tau$                                               | $2.894 \times 10^{-13}$ s | $2.903 \times 10^{-13}$ s | −0.30%    |

The two leptonic partial widths, like  $\tau_\mu$ , do not contain the electromagnetic correction of the charged lepton lines (§4.3); the nesting corrects the hadronic side only, and the residuals of the leptonic branches are retained as they are in the branching ratio and the total width.

## 4.6 $\beta$ Decay

The vector and axial-vector readouts have different standing. The vector readout runs along the flow of generation charge, and the commutator of generation-charge conservation (§2.1) protects it: insertions commute with  $\mu_2$ , the vector weight is not renormalized by the strong dwell and is identically one, which is the conservation of the vector current[12], consistent with the constancy of the  $Ft$  values of superallowed  $0^+ \rightarrow 0^+$  transitions[45]. The share remaining in the branch of the same generation is the lattice projection

$$|V_{ud}| = \sqrt{1 - \lambda^2} = 0.97443,$$

a deviation of +0.078%, with  $\lambda$  the value computed in reference [1].

The axial-vector readout has no commutator protection: total parity is a selection at the vertex and does not run along the charge flow, so the axial component is corrected by the dwell.

Step 1, cancellation in the ratio. The axial-vector weight  $g_A$  is the ratio of the axial-vector and vector readouts; on the vector side each insertion evaluates to the identity (commutator protection), and any insertion acting equally on the two readouts cancels in the ratio. The electromagnetic composite carries no lattice operator and changes no species, acting equally on both readouts: only the color composite remains in the ratio.

Step 2, the series. The axial insertions are self-energy composites on the line, both vertices falling on the dwell segment and occupying no position on the transition side, so the one-composite rule of Step 1 of §4.4 does not apply to them; the line of a nucleon is maintained by the emergent state, the dwell does not terminate, the insertions of successive layers accumulate in phase, and the series has the same form as the nesting (§4.4):

$$\sum_{n \geq 0} \left( \frac{1}{2} d_2 \alpha_s^{(0)} \right)^n = \frac{1}{1 - \frac{1}{2} d_2 \alpha_s^{(0)}} = 1.2083.$$

The same series on the vector side is the identity term by term, with sum one: the conservation of the vector current survives the series.

Step 3, vertex contact. The product of the line of a conjugate pair with an insertion member can fall on an odd direction:  $\chi_1 \bar{\chi}_4 = \chi_3$ , the landing point being the direction of the vertex itself; the configuration belongs to the vertex, and each readout has exactly one share of it. The intermediate state is colorless, color conservation restricts the configuration to the color-singlet component, and the amplitude projection is  $1/\sqrt{d_2}$ ; the coupling is taken at Born level (the evaluation rule[1]). A lepton line has no such configuration ( $\chi_2 \bar{\chi}_4 = \chi_4$ , not on an odd direction): the axial-vector weight of a lepton is one, in agreement with observation.

Step 4, assembly.

$$g_A = \frac{1}{1 - \frac{1}{2} d_2 \alpha_s^{(0)}} + \frac{\alpha_s^{(0)}}{\sqrt{d_2}} = 1.2083 + 0.0663 = 1.2746$$

The experimental value is 1.2754, a deviation of  $-0.06\%$ .

## 4.7 $\pi$ and $W$

The decay of  $\pi^\pm$  is controlled by the parity seed: the vertex for an even pairing emitting into a pseudoscalar direction vanishes identically in the massless limit, and the first nonzero term is seeded by the cycle phase of the final lepton,

$$\Gamma \propto m_\ell^2 \left( 1 - \frac{m_\ell^2}{m_\pi^2} \right)^2,$$

so helicity suppression is a corollary of parity conservation. The structure of the branching ratio  $R_{e/\mu}$ [43] is reproduced thereby; its numerical residual is amplified from the  $-0.9\%$  deviation of the framework value of  $m_\pi$ , and the improvement lies on the side of reference [1].

The nine branches of the  $W$  give the structural value  $\text{BR}(W \rightarrow \ell \nu_\ell) = 1/9 = 0.111$  (experiment 0.109). The dwell correction of the hadronic branches cannot take over the values at the  $\tau$  scale: in the framework the couplings do not run with the energy scale, and what changes with the scale is the equal-share counting of the composite shares[1]. That counting law is the remaining unfinished part of this paper; its derivation requires order-by-order counting at the amplitude level and lies beyond the scope of this paper; the evaluations of §4.5 and §4.6 do not depend on it.

## 5 The Phenomena of Decay

§1.1 to §4 have established transit, rules, records, and rates. The phenomena of decay run from the unpredictability of the instant of a single nuclear decay to the stability of atomic nuclei; five phenomena, from the simplest to the most complex, each test one link of the mechanism, and the stability of nuclei consists of the closure of transits.

## 5.1 The Exponential Decay Law

The instant at which a single nucleus decays cannot be predicted: it may decay in the next second or after a thousand years; and a nucleus that has already survived a thousand years has the same distribution of remaining lifetime as a freshly prepared one. On the ensemble side, the count falls off exponentially[5], and the rate is the only definite quantity. Memorylessness and the absence of an instant are derived from the structure below.

Memorylessness is the constancy of the loss share. The loss share is the same  $\varepsilon$  in every cycle (§3.2), and the conditional probability of surviving  $k$  further cycles after surviving  $n$  is

$$\frac{(1 - \varepsilon)^{n+k}}{(1 - \varepsilon)^n} = (1 - \varepsilon)^k,$$

independent of  $n$ : survival does not depend on the time already survived, and old samples obey the same distribution as freshly prepared ones.

The absence of an instant comes from the record criterion. The loss accumulates at the amplitude level cycle by cycle (§3.2), and accumulation is not an event; decay is completed when the loss turns into an escaping excitation carrying a record, and only when the record is established is there an event[3]. When the decay happened has no answer before the record, because before the record nothing happened. The macroscopic paradox triggered by a decay[32] dissolves accordingly: the record of the detector is the phase carried away irrevocably by a really emitted photon[3], and once the record is established everything downstream of the chain is at the probability level; what is in superposition is only the process at the amplitude level, never a macroscopic body that has been recorded.

The rate of decay is at the same time a clock of proper time. Proper time counts cycle phase,  $\Delta\tau = \Delta\Phi/m$ [3]; for a standing wave in motion the exchange share per cycle is suppressed by  $m/E$ [3], the transit sits on the formation-exchange step (§2.1), and the loss share is suppressed in the same proportion,

$$\varepsilon_{\text{motion}} = \frac{m}{E} \varepsilon, \quad \tau = \frac{E}{m} \tau_0,$$

so the lifetime read in the laboratory is dilated by  $E/m$ . Cosmic-ray muons are produced more than ten kilometers up, the distance corresponding to the rest lifetime is less than seven hundred meters, and the count reaching sea level reads precisely this dilation[22].

The lifetime is the reciprocal of the rate; the instant of decay does not exist beforehand in the state, and the magnitude of the rate is the subject of §5.4.

## 5.2 The $\beta$ Spectrum

The electron energy spectrum of  $\beta$  decay is continuous[20]. In a two-body decay the final energies are unique and the spectrum should be a single line; the measured spectrum extends instead from zero all the way to the endpoint, and the continuous spectrum once cast doubt on energy conservation itself. Pauli postulated an unobserved neutral particle for it[30], and Fermi built the theory of the contact interaction on that basis[8]. The emitted pair, the shape of the spectrum, and the endpoint are derived from the structure below.

The emitted pair comes from the pairing of parity conservation (§2.2): the emitted pair of the  $\beta$  branch is  $(e, \bar{\nu}_e)$ , and the final state is always three-body; the emitted pair of a purely leptonic transit must contain exactly one neutrino, so the energy spectrum of the muon is continuous as well. In the state space the final state of the transit is three standing waves; in observational spacetime what is read out is the energy distribution of the electron, the three-body division of energy has no

unique solution, and the spectrum is therefore continuous. The same pairing rule gives a single line for a two-body final state: the final state of  $\pi^\pm \rightarrow \mu \nu_\mu$  consists of the emitted pair alone, and the muon is monoenergetic (§4.7); the continuity of three-body and the single line of two-body spectra come from the same emitted pair. The neutrino here is what parity conservation requires of the emitted pair.

The shape of the spectrum comes from the rate formula. The strength of the transit is absorbed into the cycle rate (§4.3), the vertex is of contact form, the matrix element of an allowed transition carries no final-state momentum, and the phase space of the rate formula (§4.2) gives the whole shape of the spectrum[33]: the density of states on the electron side is  $p_e E_e$ , on the side of the massless neutrino  $(E_0 - E_e)^2$ , and the recoil balances momentum while taking almost no energy,

$$\frac{d\Gamma}{dE_e} \propto p_e E_e (E_0 - E_e)^2, \quad E_0 = \Delta m.$$

The endpoint is given by the mass difference: with  $\Delta m = 1.30 \text{ MeV}$  computed in reference [1] and  $m_e = 0.511 \text{ MeV}$ , the endpoint kinetic energy is  $0.79 \text{ MeV}$ , the experimental value being  $0.782 \text{ MeV}$ [46]. The spectrum goes to zero quadratically at the endpoint, and the mass of the neutrino distorts the shape of the approach to zero,

$$\frac{d\Gamma}{dE_e} \propto p_e E_e (E_0 - E_e) \sqrt{(E_0 - E_e)^2 - m_\nu^2};$$

the neutrino mass spectrum computed in reference [1] thereby enters the shape of the endpoint directly, the target of the endpoint measurements[29].

The particle historically inferred from the continuous spectrum is given directly by the parity structure of the vertex in the mechanism; the breaking of the same structure appears in §5.3.

### 5.3 Parity Violation

Two newly discovered mesons had the same mass and the same lifetime yet decayed to final states of opposite parity; Lee and Yang examined the literature and pointed out that parity conservation in weak processes had never been tested by experiment[10]. The next year Wu and collaborators polarized  $^{60}\text{Co}$  at a hundredth of a kelvin and found more electrons emitted opposite to the nuclear spin[11], and the asymmetry of the muon in the  $\pi \rightarrow \mu \rightarrow e$  chain was observed at the same time[37]; reflected in a mirror, the apparatus should emit more electrons along the spin, and this mirror-image experiment does not occur in nature. The helicity of the neutrino (the reading of the conserved angular-momentum component along the direction of propagation[3]) was subsequently measured to be a single left-handed one[26]. This set of facts is derived one by one from the structure below.

The mirror image in the state space is the parity reflection  $P$  (Appendix A). Among the generators that change identity, the color directions  $\chi_2, \chi_4$  are even, and electromagnetism is the elastic channel; the only odd generator is the weak direction  $\chi_3$  (§2.1). Release and elastic processes have even direction content, the evaluation preserves parity, and the strong and electromagnetic phenomena are symmetric under reflection; the transit that changes identity proceeds through the  $W$  carried by the odd direction  $\chi_3$  (§2.2), and the transit is transposed under reflection,

$$P [D_F, T_a] P = [D_F, T_{-a}], \quad P \chi_3 = -\chi_3,$$

so the mirror process is not equivalent to the original. The asymmetry appears exactly in the processes that change identity: the place of the violation coincides with the place of decay.

The violation is maximal. The combination paired with the odd component is the unique survivor, and the pairing takes the  $V-A$  form[12, 38] (§2.1); there is no second pairing to mix with,

the helicity correlation of the emitted pair is completely fixed, and the neutrino enters the emitted pair with a single helicity[26].

The coefficient of the asymmetry is fixed by the weights. The  $\beta$  asymmetry of polarized neutrons is given by the ratio of the vector and axial-vector weights, and the angular distribution of the contact matrix element (§4.2) gives the coefficient of electron emission along the nuclear spin as

$$A_n = -\frac{2g_A(g_A - 1)}{1 + 3g_A^2};$$

with  $g_A = 1.2746$  of §4.6 the tree-level coefficient is  $-0.1192$ , the experimental value is  $-0.1199$ [44], and the difference lies within the magnitude of the recoil and radiative corrections.

The change of identity and the breaking of the mirror come from one vertex.

## 5.4 The Lifetime Hierarchy

The lifetimes of unstable states fall into tiers. Strong resonances release on the scale of  $10^{-23}$  seconds, electromagnetic de-excitations on the scale of  $10^{-17}$  seconds, and weak decays extend upward from  $10^{-13}$  seconds; both being  $\pi$  mesons, the  $\pi^0$  has a lifetime of  $8.5 \times 10^{-17}$  seconds and the  $\pi^\pm$  one of  $2.6 \times 10^{-8}$  seconds, nine orders of magnitude apart[46]. The spacing of the tiers is derived from the structure below.

The first divide is the presence or absence of a transit. Release and de-excitation do not change identity, pass through no zero element, and have rates of first order in the channel coupling  $\alpha_k$  (§6); a change of identity must pass through the second-order zero-element transit (§1.1), the contact coupling of the transit  $G_F = 1/(\sqrt{2}v^2)$  (§4.3) enters the amplitude once, and the rate takes the modulus squared:

$$\Gamma_{\text{release}} \sim \alpha_k m, \quad \Gamma_{\text{decay}} \sim m \left(\frac{m}{v}\right)^4.$$

For the muon,  $(m_\mu/v)^4 = 3.4 \times 10^{-14}$ : the transit lengthens the reference scale  $1/m_\mu \sim 10^{-23}$  seconds by nearly fourteen orders of magnitude, the numerical factor of the phase space  $384\pi^3 \approx 1.2 \times 10^4$  lengthens it by four more, and together they make up the  $2.2 \times 10^{-6}$  seconds of §4.3. The decay of  $\pi^0$  emits no neutrino and is a release (§6);  $\pi^\pm$  passes through a transit and is suppressed once more by the parity seed (§4.7): the nine orders of magnitude are the direct display of the same divide on the same meson.

The second divide is the fifth power of the available energy. Within one tier, the rates are distributed with the fifth power of the available energy, the dimension of  $\Gamma \propto m^5/v^4$  being given by the rate formula (§4.3); Sargent first read the fifth power off the endpoint energies of  $\beta$  emitters[21]. Between  $\mu$  and  $\tau$ ,  $(m_\mu/m_\tau)^5 = 7.4 \times 10^{-7}$ , the observed lifetime ratio is  $1.3 \times 10^{-7}$ , and the remaining share is the branch counting of the  $\tau$  (§4.5).

The third divide is the shift of generation charge. Strange particles are produced in pairs in strong collisions yet have lifetimes of  $10^{-10}$  to  $10^{-8}$  seconds, thirteen orders of magnitude slower than the process producing them[23, 24]; Gell-Mann and Nishijima introduced a new quantum number, conserved by the strong interaction and changed by one unit in each weak decay[25, 35], and Cabibbo found that the amplitudes changing strangeness share one strength, differing only by a universal angle[13]. Strangeness in the mechanism is the generation charge (§2.1): the strong and electromagnetic vertices carry no lattice operator and cannot rewrite the generation record, so a second-generation quark can be produced only in a pair with its antiparticle, which is associated production; a change of identity has the transit as its only route, and a cross-generation change is read out further by the lattice projection of the transit as the share of the locked link,

$$|V_{us}| = \lambda = 0.2247, \quad \lambda^2 = 0.0505 \approx \frac{1}{20},$$

each generation crossed suppressing the rate by a further factor of twenty;  $\lambda$  is the value computed in reference [1], and the share  $\sqrt{1 - \lambda^2}$  remaining in the same generation has been compared in §4.6. Cabibbo's universal angle is here the universality of the lattice readout: the share belongs to the structure of the transit, whatever line passes through it.

The fourth divide is two transits. A four-vertex process of two transits in series multiplies the rate by a further fourth power of a transit; double  $\beta$  decay thereby becomes the slowest decay ever observed directly, with half-lives of the order of  $10^{20}$  years[27].  $K^0$  and  $\bar{K}^0$  are separated by two generation crossings, their mixing can proceed only by the four-vertex process of two transits (§2.2), and the smallness of the mixing follows, historically named the  $\Delta S = 2$  rule[14].

The top of the hierarchy is a rate identically zero. The muon has a single decay branch, and the electromagnetically easier  $\mu \rightarrow e\gamma$  has never appeared: if an electromagnetic vertex could rewrite the generation, the share of the radiative branch would be counted by the coupling  $\alpha$ , of order  $10^{-2}$ ; experiment has pushed the upper bound down by ten orders of magnitude to  $4.2 \times 10^{-13}$  without a single event[16], and  $\mu \rightarrow 3e$  is equally absent[17]. The electromagnetic vertex carries no lattice operator and cannot rewrite the generation record, so the amplitude of  $\mu \rightarrow e\gamma$  is identically zero; the lepton generations have no links, a lepton at the transit can only continue in its own generation, and the amplitude of  $\mu \rightarrow 3e$  is likewise zero (§2.1); the generation charge leaves with  $\nu_\mu$ ,  $e$  and  $\bar{\nu}_e$  appear as the generation-neutral emitted pair, and the path is unique (§2.2). A neutral cross-generation process has no single-transit path, historically named the absence of flavor-changing neutral currents[14]. On the quark side the same conservation law crosses generations with the share  $\lambda$ ; on the lepton side there is no link to read and the cross-generation reading is zero; the asymmetry of the two sides comes from the two constructions of the inter-generation structure[1]. A rate identically zero has no degree, and a single event suffices to refute it (§7.2); the stability of the proton is the same kind of zero (§5.5).

The spacing of the tiers is the count of transits and the shift of generation charge, and the five tiers combine into one formula,

$$\Gamma \propto m \left( \frac{m}{v} \right)^{4n} \lambda^{2|\Delta g|},$$

$n$  the number of transits,  $n = 0$  being release, and  $|\Delta g|$  the shift of generation charge; from  $10^{-23}$  seconds to  $10^{20}$  years, more than fifty orders of magnitude are given by the powers of one transit. The zero at the top is not on the tiers; it is an identity.

## 5.5 Nucleon Stability

The neutron decays with a lifetime measured in minutes; the proton is stable, with a lifetime bound of  $2.4 \times 10^{34}$  years[18]. Their masses differ by 1.4 parts in a thousand, yet the answers are opposite. The two answers and the magnitude of the neutron lifetime are derived from the structure below.

On the neutron side, the internal projection has exactly one open branch, and the mass difference  $\Delta m = 1.30 \text{ MeV} > m_e$  computed in reference [1] lets the final state be completed (§2.3). The magnitude of minutes comes from the fifth-power scaling (§5.4). With the muon as reference,

$$\frac{\Gamma_n}{\Gamma_\mu} = 96 f_0 (1 + 3g_A^2) |V_{ud}|^2 \left( \frac{m_e}{m_\mu} \right)^5,$$

where  $f_0$  is the phase-space integral in units of  $m_e$ ,  $1 + 3g_A^2$  is the combination of the vector and axial-vector readouts (§4.2), and  $g_A$  and  $|V_{ud}|$  take the values of §4.6:  $(m_e/m_\mu)^5 = 2.6 \times 10^{-12}$  lengthens the lifetime by twelve orders of magnitude,  $96 f_0 (1 + 3g_A^2)$  shortens it again by about three, and together they give the order of  $10^3$  seconds, the observed order of minutes[46]. The electron forms at, and leaves, the charged emergent state; the electromagnetic correction of the

electron line is not derived in this paper, and the precise value of the lifetime is not among the outputs of this paper.

On the proton side, the branches in which a quark becomes a quark have heavier final states and are closed by the completion condition; the branches in which a quark becomes a lepton or a neutrino are closed by charge (§2.3), the charge difference being fractional while channel excitations carry only integer charge, so no emitted pair can balance it. The latter is the zero of an identity, with no degree. In the state space all branches of the proton are closed; in observational spacetime there is no event. The stability of every atomic nucleus rests on this zero and on the completion condition.

Two adjacent states on the spectrum, one with its branch just open, the other with all branches closed; the divide is only 1.4 parts in a thousand.

The five phenomena come from one structure. The exponential decay law gives the boundary of records, the  $\beta$  spectrum gives the emitted pair, parity violation gives the breaking on the unique odd direction, the lifetime hierarchy gives the powers of the transit and the shift of generation charge, and nucleon stability gives the two faces of completion and closure. Every quantitative reading comes from the evaluation at the same vertex; the definition of decay has withstood the tests from the release at  $10^{-23}$  seconds to the double  $\beta$  decay at  $10^{20}$  years.

## 6 The Classification of Decays

In standard usage every spontaneous passage of an unstable state to a lower state is called a decay, and decays are divided by interaction into strong, electromagnetic, and weak. The preceding sections have established the class that changes identity, corresponding to weak decay. The three classes are distinguished within one cycle structure: each class has its own source of loss, and the three share the loss share per cycle and the readout structure of §3.

### 6.1 Three Classes of Decay

The amplitude of an unstable state is lost cycle by cycle. Of the three steps of the cycle, the formation-exchange step trades amplitude with the background and the pattern-reconstruction step fixes the position (§1.1); the exchanged amplitude either merges into the background through the zero element or falls back along a channel, and the reconstruction either stays in place or moves beyond a barrier. Through the zero element, back along a channel, or beyond a barrier: these are the only three sources of spontaneous loss.

The first class is zero-element transit. The amplitude of the initial state merges into the background through the zero element, the final state re-forms from the background, identity changes, and the emitted pair carries the record away (§2); the transit is a second-order process, and its rate carries the suppression  $(m/v)^4$  (§4). All weak decays belong to this class;  $\beta$  decay and the decays of  $\mu$ ,  $\tau$ ,  $K$  mesons,  $\pi^\pm$ , and  $W$  are its instances.

The second class is resonance release. A channel excitation returns its weight to the channel, the identity of the standing wave is unchanged, and no zero element is passed; the strength of the fall-back is of first order in the channel coupling  $\alpha_k$ , given by the resonance structure of the vertex direction  $k$ [1]. Strong and electromagnetic decays belong to this class;  $\rho \rightarrow \pi\pi$ ,  $\pi^0 \rightarrow \gamma\gamma$ , the  $\gamma$  de-excitation of nuclei, and the decay of  $Z$  are its instances.

The third class is **barrier rearrangement**. Constituents cross a barrier as a whole along the multiplicative direction and are rebuilt on the far side; the identity of each constituent is unchanged,



and no zero element is passed; the crossing probability is suppressed by  $e^{-\beta d}$  with  $d$  the metric width of the barrier[3].  $\alpha$  decay and spontaneous fission belong to this class.

| Source of loss        | Zero element | Identity  | Final-state $\nu$ | Loss per cycle | Instances                                                                            |
|-----------------------|--------------|-----------|-------------------|----------------|--------------------------------------------------------------------------------------|
| zero-element transit  | yes          | changed   | possible          | $(m/v)^4$      | $\beta, \mu, \tau, K, \pi^\pm, W$                                                    |
| resonance release     | no           | unchanged | none              | $\alpha_k$     | $\rho \rightarrow \pi\pi, \pi^0 \rightarrow \gamma\gamma, \gamma$ de-excitation, $Z$ |
| barrier rearrangement | no           | unchanged | none              | $e^{-\beta d}$ | $\alpha$ decay, spontaneous fission                                                  |

The three classes are distinguished observationally by the neutrino. The neutrino is not in the catalogue of record carriers (§3.1); its appearance in a final state has only one source, the emitted pair of a zero-element transit; a process emitting a neutrino is necessarily a change of identity, and the final states of resonance release and barrier rearrangement contain no neutrino.

The three classes share one readout structure. The loss share per cycle  $\varepsilon$  enters the readings of §3.3; the exponential decay law, the Breit–Wigner line shape, and the branching ratios hold alike for the three classes, and the difference lies only in the source of  $\varepsilon$ :

$$\varepsilon_{\text{transit}} \propto \left(\frac{m}{v}\right)^4, \quad \varepsilon_{\text{release}} \propto \alpha_k, \quad \varepsilon_{\text{rearrangement}} \propto e^{-\beta d}.$$

The fast end of the lifetime hierarchy (§5.4) is resonance release and the slow end is zero-element transit; barrier rearrangement spans the range between them, its exponential dependence making its lifetime vary by orders of magnitude with the energy difference. The rates of the latter two classes are given in the following two subsections.

## 6.2 Resonance Release

Resonance release is the fall-back of a channel excitation. Resonances and excited states are channel excitations on top of a stable standing wave; the convolution operator  $K$  is diagonal in the character basis, and the weight of the excitation flows back along the same channel, passing through no zero element and changing the identity of no standing wave (§2.2). The fall-back proceeds once per cycle, and the final state is an excitation of the channel itself: the strong channel gives hadrons, and the electromagnetic channel gives photons.

The rate of release is given by a first-order vertex. The coupling of the channel is  $\alpha_k = |R_k|^2/N_k^{\text{eff}}[1]$ , and the matrix element of the fall-back contains a single vertex,

$$|M|^2 = c \alpha_k m^2,$$

with  $m$  the mass of the excitation and  $c$  the structure factor of angular momentum and isospin; inserted into the golden rule in continuous time (§4.2) with the two-body phase space  $\rho_2 = p_f/(8\pi m^2)$ ,  $p_f$  the final-state momentum,

$$\Gamma_{\text{release}} = |M|^2 \rho_2 = \frac{c \alpha_k}{8\pi} p_f \sim \alpha_k m, \quad \varepsilon_{\text{release}} = \frac{\Gamma_{\text{release}}}{v} \sim \alpha_k \frac{m}{v}.$$

The matrix element of a zero-element transit contains the contact coupling  $G_F = 1/(\sqrt{2}v^2)$ , and the rate carries  $(m/v)^4$  on squaring (§5.4); the matrix element of a release contains no contact coupling, the generation charge does not shift, and the combined formula of the lifetime hierarchy gives the same result at  $n = 0$ . The widths of strong resonances are thereby measured in hundreds of MeV; the ratio of width to mass of  $\rho \rightarrow \pi\pi$  is 0.19, of the same order as  $\alpha_s^{(0)} = 0.115$ ; the

electromagnetic coupling  $\alpha_{\text{EM}}^{(0)} = 0.0079$  is smaller by more than an order of magnitude, the widths of  $\gamma$  de-excitations are suppressed accordingly, and  $\pi^0 \rightarrow \gamma\gamma$  contains two electromagnetic vertices and is of second order in  $\alpha_{\text{EM}}$ .

The release of the  $Z$  is an instance of the same type.  $Z$  belongs to an even direction and is a channel excitation (§2.2); its width adds term by term over the open directions,

$$\Gamma_Z = \sum_f \Gamma(Z \rightarrow f\bar{f}), \quad \Gamma_{\text{invisible}} = N_\nu \Gamma(Z \rightarrow \nu\bar{\nu}),$$

and the invisible share counts the number of neutrino generations  $N_\nu = (p-1)/2 = 3[1]$ , the LEP count being  $N_\nu = 2.984 \pm 0.008[28]$ . The width of a release counts directions and carries no transit suppression; the rate of barrier rearrangement comes from yet another source.

### 6.3 Barrier Rearrangement

Barrier rearrangement is a displacement of position as a whole. A part of a composite body, an  $\alpha$  cluster or a fission fragment, moves as a whole from inside a barrier to outside it; the standing wave and identity of each constituent are unchanged, and only the position changes. Reconstruction inside the barrier region is suppressed, and site-by-site motion along the additive direction cannot pass; one multiplicative step does not pass site by site but connects directly two positions differing by a factor  $p$ , the suppression does not act on it, and the pattern is rebuilt directly on the far side of the barrier[3].

The rate of rearrangement is given by the metric width of the barrier. In the metric  $d(x, y) = |\log(x/y)|$  the KMS weight  $n^{-\beta} = e^{-\beta d(1, n)}$  is the exponential of the metric distance[1]; one multiplicative step  $x \mapsto px$  crosses the metric distance  $\log p$  with weight  $p^{-\beta}$ ; the amplitude share passing through the multiplicative channel per cycle is suppressed by the metric width  $d$  of the barrier region, and the probability that the event lands on the far side is the modulus squared by the Born rule[3],

$$A \propto e^{-\beta d/2}, \quad \varepsilon_{\text{rearrangement}} = C e^{-\beta d}, \quad \Gamma = v C e^{-\beta d}, \quad \log \tau = \beta d - \log(vC),$$

with  $C$  the reconstruction factor in front of the barrier. The metric width of the barrier grows as the available energy  $E$  decreases; for a Coulomb barrier (the potential given in reference [4]) the exponent of the crossing varies with  $Z/\sqrt{E}$ , in agreement with Gamow's result[6], so the logarithm of the lifetime is linear in  $Z/\sqrt{E}$ , the Geiger–Nuttall law[7]: the  $\alpha$  half-lives of a family of nuclides extend from below a microsecond to  $10^{19}$  years, more than thirty orders of magnitude corresponding to an energy difference of a few MeV. This span comes from the exponential dependence and belongs to a source different from the transit powers of §5.4; spontaneous fission is likewise a rearrangement of position.

The derivation of the crossing is established in the mechanism of motion[3], and this paper takes its result. The loss shares of the three classes enter one readout; the numbers and predictions of this paper are collected in the next section.

## 7 Conclusion

The numerical outputs and structural corollaries of this paper are compared with experiment item by item, and the amplitudes identically zero give falsifiable predictions.

## 7.1 Numerical Summary

The numerical outputs compared with experiment are as follows; all inputs on the side of this paper are computed outputs of reference [1], and the experimental column is taken from reference [46] and the individual experiments.

| Quantity                                    | This paper               | Experiment               | Deviation |
|---------------------------------------------|--------------------------|--------------------------|-----------|
| $G_F [\text{GeV}^{-2}]$                     | $1.16665 \times 10^{-5}$ | $1.16638 \times 10^{-5}$ | +0.023%   |
| $\tau_\mu [\text{s}]$                       | $2.18709 \times 10^{-6}$ | $2.19698 \times 10^{-6}$ | -0.45%    |
| $\Gamma(\tau \rightarrow e) [\text{GeV}]$   | $4.050 \times 10^{-13}$  | $4.040 \times 10^{-13}$  | +0.25%    |
| $\Gamma(\tau \rightarrow \mu) [\text{GeV}]$ | $3.939 \times 10^{-13}$  | $3.943 \times 10^{-13}$  | -0.10%    |
| $\text{BR}(\tau \rightarrow \mu)$           | 0.1732                   | 0.1739                   | -0.40%    |
| $\tau_\tau [\text{s}]$                      | $2.894 \times 10^{-13}$  | $2.903 \times 10^{-13}$  | -0.30%    |
| $ V_{ud} $                                  | 0.97443                  | 0.97367                  | +0.078%   |
| $g_A$                                       | 1.2746                   | 1.2754                   | -0.06%    |

The table contains no inherited correction. The electromagnetic correction of the charged lepton lines is not derived in this paper;  $\tau_\mu$  and the leptonic branches of the  $\tau$  do not contain this share, which in the standard calculation is -0.42% (§4.3). Beyond the numbers, there are nine points of agreement at the structural level:

- the exponential decay law is a corollary of the cycle readout (§3.3);
- the Breit-Wigner line shape is the resolvent of the sum over cycles (§3.3);
- the  $V-A$  pairing is given by parity conservation (§2.1)[12];
- that a charged current must change the quark species is a corollary of the zero-element reflection (§2.1);
- the prohibition of flavor-changing neutral currents is given by the conservation of generation charge (§2.1);
- the conservation of the vector current and the constancy of  $Ft$  are given by the commutator of generation charge (§4.6);
- that the axial-vector weight of leptons is one is given by the absence of the contact configuration (§4.6);
- the visible and invisible classes of the  $Z$  width are given by the evenness of the channel excitation (§2.2);
- the decoupling of the neutrino from the photon bath is given by electric neutrality (§2.3).

## 7.2 Testable Predictions

The following amplitudes are identically zero under the mechanism of this paper, and the observation of any one of them refutes it:

- $\mu \rightarrow e\gamma$  (conservation of generation charge);
- $\mu \rightarrow 3e$  (no links in the inter-generation structure of leptons);

- proton decay (closure by charge);
- $n-\bar{n}$  oscillation (all colored directions are complex characters)[19].

Two derivations remain to be completed. First, the electromagnetic correction of the charged lepton lines: once it is derived, the  $-0.45\%$  of  $\tau_\mu$  should disappear. Second, the hadronic branches of the  $W$ : once the equal-share counting law is established,  $\text{BR}(W \rightarrow \ell\nu)$  becomes a numerical output of the framework, and the correction should move the structural value  $1/9$  toward the observed  $0.1086$ .

Two phase structures lie outside the scope of this paper. CP violation in the cross-generation phases (the  $K$  and  $B$  meson systems[39]) and the strong CP problem belong to the part of the generation-structure phases not yet completed; the phase inputs of this paper stop at  $\delta_2$  and  $\lambda$  of reference [1], two real inputs cannot carry a CP phase angle, and the boundary of this paper is drawn here. The Majorana nature of the neutrino[34] and the criterion for neutrinoless double  $\beta$  decay are among the predictions of reference [1] and are not repeated here.

### 7.3 Concluding Remarks

Decay is a change of identity and proceeds through zero-element transit. The first-order matrix element for changing identity is identically zero and the second-order one is in general nonzero, so identity cannot be rewritten directly and can only merge into the zero element first and re-form from the background afterward. The transit is equivalent for all directions, and all selection comes from the evaluation at the two vertices: the three conservation laws of electric charge, generation charge, and parity restrict the change of identity to the two kinds  $d \leftrightarrow u$  and  $\ell \leftrightarrow \nu_\ell$ , and the states with all branches closed are stable matter.

A process becomes an event in observational spacetime only through a record. The record of a decay is the difference of identity, carried by the emitted standing waves, and the transit is complete when the final state is completed and the carrier has escaped. The position and width of the spectral line and the count ratio of the branches come from one place: the former two are the phase and modulus of the cycle eigenvalue, the latter the distribution of the loss share.

Decay runs in one direction only, because the zero element absorbs and does not release, the occupation after the loss does not rise again, and the escape of the carrier cannot be undone. The rate is converted from the loss share per cycle, and the lifetime hierarchy of more than fifty orders of magnitude comes from the powers of one transit; the Fermi constant, the lifetimes of  $\mu$  and  $\tau$ , the branching ratios,  $|V_{ud}|$ , and the axial-vector weight are given thereby, with all inputs quantities fixed beforehand in reference [1].

The decay defined in this paper is the change of identity, corresponding to weak decay in the standard classification; strong decay, electromagnetic decay, and  $\alpha$  decay are, in this framework, resonance release and barrier rearrangement, which do not change identity; the three classes share one readout structure and differ in the source of loss. With this, the mechanisms of the formation, motion, thermalization, and decay of matter have each been established on the same state space[1, 3, 2]. The four processes share one cycle: formation builds the standing wave in the cycle, motion moves the reconstruction center with the cycle, thermalization relaxes the occupations of many standing waves through the zero element, and decay replaces the identity of a single standing wave through the zero element. They are four ways in which one cyclic exchange proceeds, and their unified dynamics is the work to come.

### Data Availability

No data sets were generated or analyzed in this study; all results follow from analytic derivation.

## Conflict of Interest

The author declares no conflict of interest.

## A Verification of Parity Conservation

The operator content and the process-by-process check of parity conservation (§2.1) are as follows.

The parity reflection  $P$  is defined by  $(Pf)(x) = f(-x)$ . Characters are eigenfunctions of  $P$ , and multiplication operators and rank-one operators are accordingly eigenoperators of  $P$ :

$$P\chi = \chi(-1)\chi, \quad P M_\chi P = \chi(-1) M_\chi, \quad P |\chi\rangle\langle\chi'| P = \chi(-1)\chi'(-1) |\chi\rangle\langle\chi'|,$$

the zero element is fixed ( $P\delta_0 = \delta_0$ ), and the transit matrix element satisfies

$$P [D_F, T_a] P = [D_F, T_{-a}].$$

The background is invariant under  $P$ , and  $P$ -odd operators evaluate to zero; the total parity of a vertex is the product of the eigenvalues of its factors, which is the product rule for direction parity. The availability of record components (lepton records purely  $\iota$ -even, quark records containing an  $\iota$ -odd component) comes from the way conjugate pairs carry species (§2.1): a record distinguishing two species must distinguish the two branches  $\chi$  and  $\bar{\chi}$ , and this component changes sign under  $\iota$ .

Taking the transit of the muon as an example, the three conservation laws are checked item by item:

$$\mu^- \rightarrow \nu_\mu + (e^-, \bar{\nu}_e) : \quad \underbrace{(+)(-)\times(+)(-)}_{\text{parity}} = +, \quad \underbrace{g(\nu_\mu) = g(\mu)}_{\text{generation charge}}, \quad \underbrace{-1 = 0 + (-1 + 0)}_{\text{charge}},$$

the line continues in its generation, the emitted pair is generation-neutral, and the path is unique.

For each transit the direction parity and the number of quark-side participants are listed, and the verdict of conservation agrees with observation:

| Process                                    | Direction parity | Quark participants | Verdict                                               |
|--------------------------------------------|------------------|--------------------|-------------------------------------------------------|
| $\mu \rightarrow \nu_\mu (e, \bar{\nu}_e)$ | +                | 0                  | allowed; exists                                       |
| $d \rightarrow u (e, \bar{\nu}_e)$         | −                | 1                  | allowed; exists                                       |
| $\tau \rightarrow \nu_\tau (q, \bar{q}')$  | −                | 1                  | allowed; exists                                       |
| $s \rightarrow u (d, \bar{u})$             | +                | 2                  | allowed; exists                                       |
| $s \rightarrow u (e, \bar{\nu}_e)$         | −                | 1                  | allowed; exists                                       |
| $\mu \rightarrow \nu_\mu (e^+, e^-)$       | −                | 0                  | forbidden; absent                                     |
| $\mu \rightarrow \nu_\mu (\nu, \bar{\nu})$ | −                | 0                  | forbidden; absent                                     |
| $\mu \rightarrow e (e^+, e^-)$             | +                | 0                  | parity-allowed; forbidden by generation charge (§2.1) |
| emitted pairs of type $(q, \bar{\ell})$    | —                | —                  | forbidden by charge conservation                      |

## B Configurations of the Dwell Correction

The configuration count of the dwell correction (§4.4). The insertion pair is  $\{\chi_2, \chi_4\}$  (mutually conjugate single-factor generators); both lines of the emitted pair are quark species. The configurations of each composite are the combinations (member, line), four in all; the intermediate residence is the position of the line multiplied by the conjugate of the emitted member. Taking the  $(d, \bar{u})$  pair (both lines read out in the  $\chi_5$  direction) as an example:

| Member   | Line   | Intermediate residence                         | Contribution |
|----------|--------|------------------------------------------------|--------------|
| $\chi_4$ | line 1 | $\chi_1$ (colored)                             | nonzero      |
| $\chi_2$ | line 1 | $\chi_3$ (color-neutral, electrically neutral) | zero         |
| $\chi_4$ | line 2 | $\chi_1$ (colored)                             | nonzero      |
| $\chi_2$ | line 2 | $\chi_3$ (color-neutral, electrically neutral) | zero         |

The normalization is shared equally over the unitary displacement algebra, each of the four configurations taking one quarter; the vertex coupling of a neutral intermediate residence is zero (the generalization of the electric-neutrality exemption), and the nonzero configurations total one half, the share of Step 3 of §4.4. When the same count is applied to the Cabibbo correction of reference [1], the intermediate residences of both configurations carry charge ( $\chi_1$  and  $\chi_5$ ), the nonzero configurations total one, in agreement with the coefficient there; applied to the electrically neutral neutrino, all configurations are zero, returning to the electric-neutrality exemption itself. The three cases come from one rule. Every layer of the nesting (Step 5 of §4.4) repeats the same count: the object of the recursion is the segment, and the configuration space of a segment does not change with the layer.

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